

Freshman Financial Shock after the Great Recession

1. Introduction

Attending college is a major economic commitment for any household. Educated young people spend years foregoing earnings or accumulating debt to improve their future labor market outcomes. When households face severe wealth shocks, like the 2008 housing crisis, this decision becomes even more difficult. This paper asks whether a macroeconomic crisis affects the financial, labor supply, and human capital plans of students entering higher education.

Though the 2008 Great Recession influenced the entire nation, its severity varied continuously across states. Some states, including California and Florida, experienced tremendous home value losses and unemployment spikes; others, including North Dakota and Texas, with more of their economies concentrated in agriculture or energy, were comparatively insulated. This paper measures that variation directly, using each state's 2007-to-2009 change in its annual-average unemployment rate as a continuous shock, rather than sorting states into a small number of treatment and control groups. Even if a student attends college out-of-state, an entering freshman's expectations are primarily formed by their economic experience at home.

The analysis uses the HERI Freshman Survey, which captures students' backgrounds, attitudes, and goals at the point of college entry. Two outcomes are examined: financial anxiety (whether the student is concerned about their ability to afford college) and anticipated labor supply (whether they expect to work to help pay for their education).

To isolate the causal impact of the crisis, controls are implemented for student background, including race, high school GPA, first-generation college student status, and parental income. Institutional selectivity is also directly controlled for. Accounting

for these differences allows students from states with different degrees of recession exposure to be compared directly.

The results show no significant differential effect of a state's recession severity on either outcome. Financial anxiety shows no significant response once background controls are included ($\beta_3 = -0.0011$, $p = 0.421$). For anticipated labor supply, the coefficient is similarly small and statistically indistinguishable from zero ($\beta_3 = +0.0007$, $p = 0.735$), and this null result holds across five alternative ways of defining the treatment variable. An event-study extension shows no differential movement in either outcome in any year before or after the recession, and formal tests for heterogeneity by parental income and first-generation status find no gradient in the recession's effect. Together, these results suggest that state-level recession severity did not have a robust differential effect on how entering freshmen perceived their financial situation or planned to respond to it.

2. Literature Review: Labor Shocks & Human Capital

Identifying a causal effect requires a design that accounts for pre-existing differences across households. Angrist and Evans (1998) provide a clear example of this approach, using sibling sex composition as an instrument for family size. This study uses a similar logic, exploiting continuous variation in how hard the housing crash hit different states. Students from more severely affected states should show stronger responses in financial concern (`fin_concern`) and plans to work (`work_pay`), in proportion to the size of their state's shock.

A negative wealth shock can also change how people think about time and work. Camerer et al. (1997) studied New York City cab drivers who set a fixed daily earnings target. Students face an analogous situation: they need to earn enough to cover fixed tuition costs. Under this logic, a drop in household income should push students to

work more. However, if the local job market collapses at the same time, that plan may not be feasible.

Wealth shocks also affect academic outcomes directly. Carter, Greenberg, and Walker (2017) show that cuts to a student's material resources lower academic performance. This study looks at the same mechanism from the other direction: for freshmen entering after 2008, family wealth losses should shape decisions made before college begins. High school GPA (HSGPA) is used as a balance check to confirm that students from more- and less-affected states were academically similar before the shock hit.

Students also adjust their academic plans based on what they see happening around them. When local labor markets collapse, students face real uncertainty about which fields will actually lead to jobs—shaping decisions made before they set foot on campus. Though major choice cannot be causally identified here, this behavioral channel helps explain why financial anxiety and labor supply expectations are the right outcomes to study: both are measured at the moment of college entry, before any coursework or campus experience can shape them.

The welfare consequences of these responses depend on whether they reflect sound planning or fear-driven choices. Bleemer and Mehta (2022) argue that shifting toward higher-earning fields during a downturn reflects smart investment—students reading the labor market correctly and adjusting accordingly. Martorell and Clark (2014) offer a counterpoint: credentials carry returns even when little skill is actually gained, meaning that individually rational responses to a recession may not always reflect genuine human capital investment. Because the labor-supply result in this analysis is null, this tension remains an open question.

A key design choice here is measuring recession severity continuously across nearly every state, rather than sorting states into a small number of treatment and control groups. Comparisons based on only a handful of units create a well-known statistical problem: with few clusters, standard cluster-robust inference is unreliable, and remedies such as a wild cluster bootstrap (Cameron, Gelbach & Miller, 2008) are often necessary. This analysis instead follows Yagan (2019), who measures Great Recession severity with each area's continuous change in unemployment rate. Using this continuous measure across nearly every state provides close to fifty clusters, which is enough for ordinary cluster-robust inference to be reliable (Cameron & Miller, 2015).

3. Theoretical Framework: Wealth Shocks & Time Budgets

We can model these dynamics through a standard household allocation framework. A student maximizes utility over consumption C and leisure L :

$$\max_{C,L} U(C, L)$$

This choice is subject to a standard budget constraint:

$$C = w(T - L) + I$$

where w is the part-time wage, T is the total time endowment, $(T - L)$ represents continuous hours worked, and I is the parental transfer subsidizing college attendance.

The recession operates directly through I , in proportion to the severity of the local labor market shock. Rather than a fixed shift applied to a small set of treatment states, ΔI varies continuously with the size of each state's 2007–2009 unemployment increase shock_s. Under basic labor supply theory, a larger shock should push students to work more ($\Delta(T - L^*) > 0$ as shock rises) to fill the gap. The binary variable `work_pay` measures this intent on the extensive margin.

A separate prediction concerns financial anxiety and income: because poorer families have less of a buffer against a wealth shock, anxiety at college entry should be strongly related to parental income regardless of the recession. Whether the recession's differential effect itself varies by income is a distinct, testable question, addressed directly in Section 6.

For work expectations, a larger shock should raise planned work through the standard income effect, unless local job markets are simultaneously too weak to support that plan, in the spirit of Camerer et al. (1997). The empirical analysis evaluates which of these channels dominates using a large and representative set of states.

4. Data and Sample Construction

The dataset is constructed by merging four SPSS modules from the HERI Freshman Survey using the keys SUBJID and YEAR, plus one external source:

- **DEMOGRAPHICS.SAV:** Contains STATE, YEAR, SEX, RACEGROUP, INCOME, and FIRSTGEN.
- **FUNDS.SAV:** Contains FINCON, SELECTIVITY, and STUDWGT.
- **PLANS.SAV:** Contains FUTACT12.
- **HIGHSCHOOL.SAV:** Contains HSGPA.
- **State-level unemployment rates:** Annual-average unemployment rates (BLS Local Area Unemployment Statistics, via FRED), used to build a time-invariant shock measure: $\text{shock}_s = (\text{2009 rate}) - (\text{2007 rate})$ for each state s .

Before merging, duplicate student-year records in each module are resolved by keeping the first observation. The DEMOGRAPHICS module is treated as the authoritative source for the cohort year variable when conflicts arise.

The sample includes every state with a matched unemployment-rate series—49 states in total (Delaware and Wyoming have no participating HERI institutions in this extract's sample years and are absent; all other states and DC are represented). The cohort years used are 2003–2007 and 2009–2010, allowing pre-trends to be tested across five years. The 2008 cohort is dropped to avoid contamination from the acute crisis period. Of 2,513,524 merged student-year records, 2,009 rows come from states without a matched unemployment series and are dropped, leaving a final sample of 2,511,515 student-year observations.

The key variables are built as follows:

- `shock_s` is the continuous 2007-2009 change in the annual-average unemployment rate of state `s`
- `post_t` is 1 if the cohort entered in 2009 or later, and 0 otherwise.
- The outcome Y^1 (Financial Anxiety, stored as `fin_concern`) comes from FINCON. It is coded 1 if FINCON equals 3 ("Major Concern") and 0 otherwise.
- The second outcome Y^2 (Work Expectation, stored as `work_pay`) is derived from FUTACT12, which asks entering freshmen how likely they are to work to help pay for college. Responses follow a four-point scale: 1 = Very likely, 2 = Somewhat likely, 3 = Not sure, 4 = Not likely. It is coded as a binary indicator equal to 1 if the student responds "Very likely" or "Somewhat likely," and 0 if they respond "Not sure" or "Not likely." This coding treats uncertainty as a non-commitment to work.

Controls include continuous HSGPA, binary SEX (stored as `gender`), a 7-level RACEGROUP factor (stored as `race`), continuous institutional SELECTIVITY, binary FIRSTGEN status, and a 15-category INCOME factor. Observations missing control variables are dropped via listwise deletion, R's default handling for complete-case regression estimation; per-column sample sizes in Table 2 reflect this.

Table 1 summarizes the raw relationship between period and each outcome.

Period	Mean state_shock	Fin. Anxiety	Work to Pay
Pre-recession (2003-2007)	4.50	0.122	0.200
Post-recession (2009-2010)	4.55	0.114	0.178

Table 1: Raw outcome means by period. N = 1,970,500 pre-recession; 541,015 post-recession.

Both outcomes fall slightly from the pre- to post-recession period in the raw data. This reflects a long-run downward drift across the 2003–2010 window that is unrelated to the recession specifically. This is why the shock x post interaction, not the raw post coefficient, is the relevant causal estimate: it asks whether more-exposed states moved differently from less-exposed states.

5. Empirical Strategy

We estimate the causal impact of the housing crash using an ordinary least squares (OLS) linear probability model:

$$Y_{ist} = \beta_0 + \beta_1 shock_s + \beta_2 post_t + \beta_3 (shock_s \times post_t) + X'_{ist} \gamma + \varepsilon_{ist}$$

where i indexes the student, s indexes the state, and t indexes the year. X'_{ist} is the vector of student background and institutional controls.

Each parameter has the following interpretation:

- β_1 captures the baseline (pre-recession) relationship between a state's eventual recession severity and the outcome.

- β_2 captures national time trends shared by all students.
- β_3 is the Difference-in-Differences estimate of interest: the effect of one additional percentage point of state-level unemployment increase on the outcome, post- versus pre-recession.

Standard errors are clustered by state. With close to fifty state clusters, ordinary cluster-robust inference is reliable (Cameron & Miller, 2015). Because shock_s is time-invariant at the state level, state fixed effects would absorb shock_s; the baseline model relies on individual controls X_{ist} to absorb compositional differences across states.

Two additional specifications extend the main equation:

1. **Event Study:** Replaces post_t with a full set of year indicators interacted with shock_s, relative to 2007 as the omitted base year:

$$Y_{ist} = \alpha + \sum_{t \neq 2007} \delta_t (\text{shock}_s \times 1[\text{year}=t]) + X'_{ist} \gamma + \text{year FE} + \epsilon_{ist}$$

2. **Income Subgroups:** The main specification is re-estimated separately within each parental-income bracket to test directly whether β_3 varies by income.

6. Empirical Results & Sensitivity Checks

Financial Anxiety

Without controls, the shock_s x post_t coefficient is small and statistically indistinguishable from zero ($\beta_3 = 0.0026$, $p = 0.255$). Adding controls flips its sign but not its significance ($\beta_3 = -0.0011$, $p = 0.421$). The no-controls post coefficient is negative and significant ($\beta_2 = -0.0201$, $p = 0.040$), reflecting the broad downward trend in anxiety over the sample period rather than a differential recession effect.

Work Expectation

Without controls, $\beta_3 = -0.0033$ ($p = 0.406$); with full controls, $\beta_3 = +0.0007$ ($p = 0.735$). Neither estimate is statistically significant, and the sign flips across specifications, consistent with sampling noise around zero.

Variable	(1) Fin. Anxiety [No Controls]	(2) Fin. Anxiety [Full Controls]	(3) Work Exp. [No Controls]	(4) Work Exp. [Full Controls]
shock _s (β_1)	0.0029 (0.0032)	0.0010 (0.0017)	0.0012 (0.0063)	-0.0009 (0.0060)
post _t (β_2)	-0.0201* (0.0095)	-0.0014 (0.0069)	-0.0074 (0.0186)	-0.0304* (0.0122)
shock $\times$ post (β_3)	0.0026 (0.0023)	-0.0011 (0.0013)	-0.0033 (0.0039)	0.0007 (0.0021)
Controls	No	Yes	No	Yes
Clusters (states)	49	49	49	49
N	2,412,035	2,052,827	2,305,891	2,091,503

Table 2: Main Difference-in-Differences Coefficients, continuous state shock. Standard errors clustered by state (stata/CR1 type) in parentheses. * $p < 0.05$.

The gap between no-controls and full-controls N reflects listwise deletion on controls: 359,208 rows drop out of the anxiety model and 214,388 out of the work model due to missing control data. Reporting both specifications side-by-side demonstrates that complete-case listwise deletion does not alter the main findings.

Alternative Functional Forms

To test robustness, five treatment specifications are estimated for each outcome:

Specification	Financial Anxiety (β_3)	Work Expectation (β_3)
Continuous shock (baseline)	-0.0011 (p = 0.421)	+0.0007 (p = 0.735)
Median split	-0.0017 (p = 0.626)	-0.0031 (p = 0.674)
Extreme terciles only	-0.0054 (p = 0.268)	+0.0028 (p = 0.632)
Quadratic shock	+0.0037 (p = 0.663)	-0.0018 (p = 0.921)
Top-quartile threshold	-0.0047 (p = 0.104)	+0.0042 (p = 0.466)

Table 3: Robustness of the shock $\times$ post coefficient across five functional-form specifications.

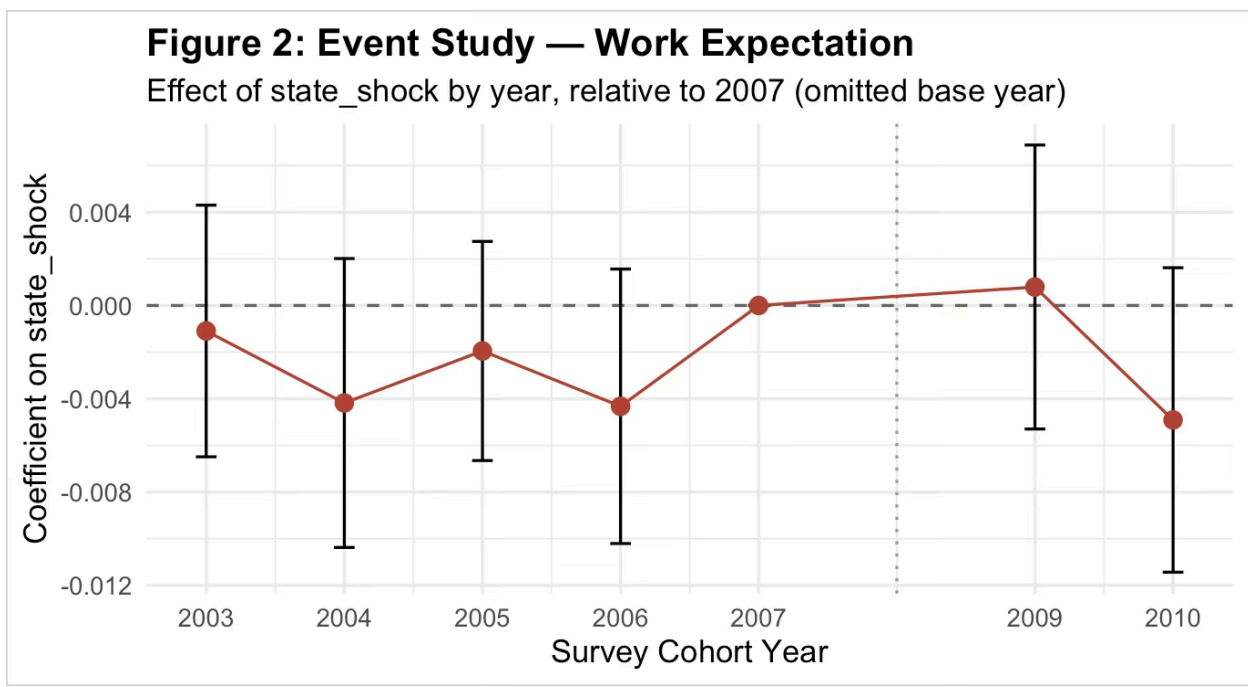
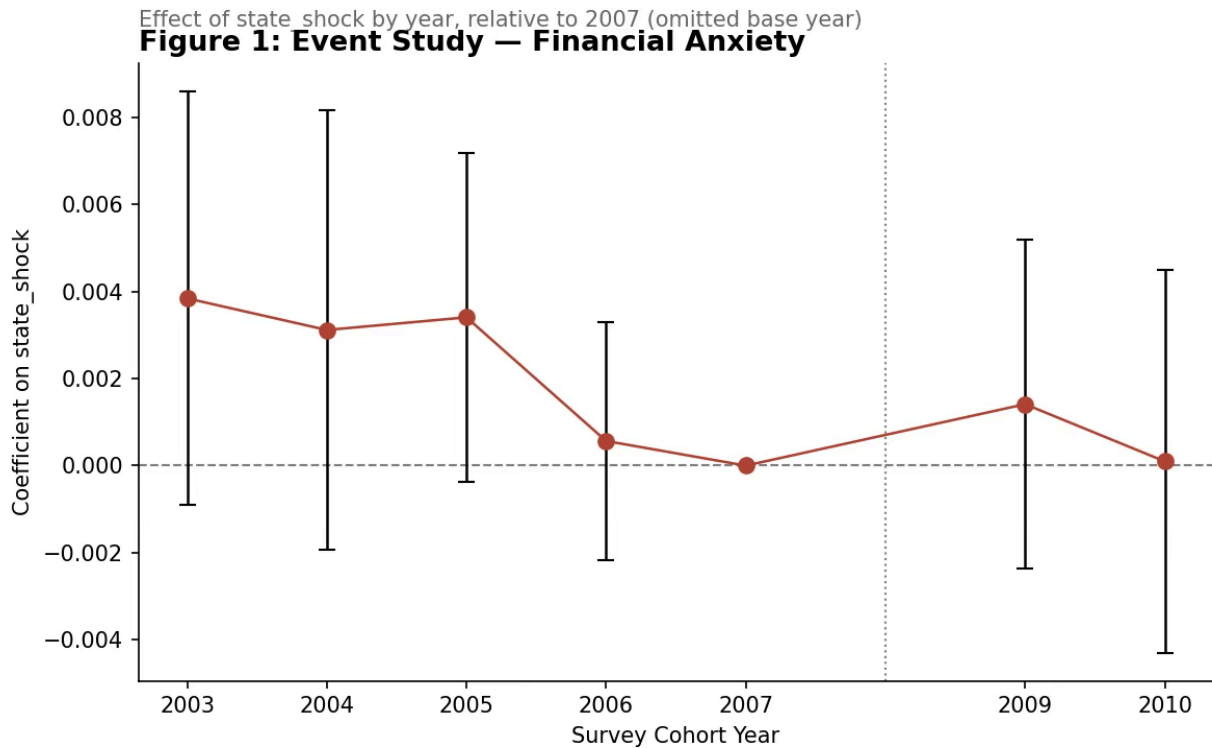
All five specifications yield null results for both outcomes.

Event Study

Figures 1 and 2 plot annual coefficients on shock_s relative to 2007. Pre-period coefficients (2003–2007) are flat and insignificant for both outcomes (ranging from

+0.0006 to +0.0038 for anxiety and -0.0043 to -0.0011 for work expectations).

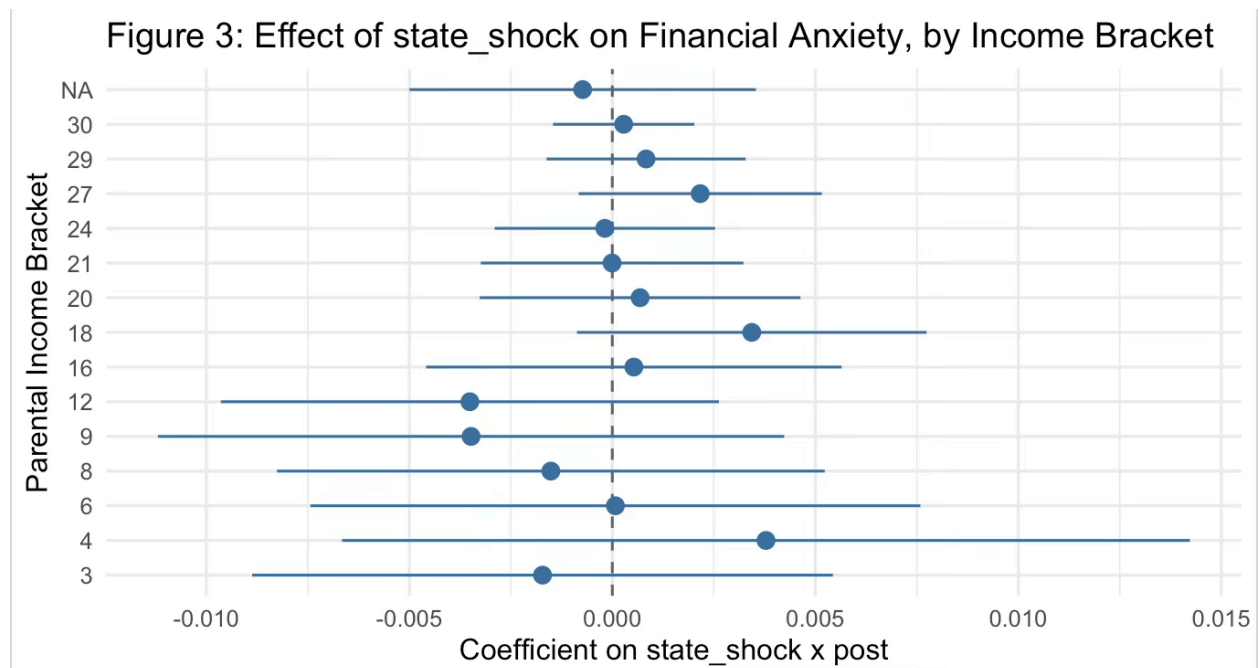
Post-period coefficients (2009–2010) are similarly indistinguishable from zero.



Heterogeneity by Parental Income & First-Generation Status

Parental income is strongly related to baseline financial anxiety: full-control model coefficients range from -0.03 at the second-lowest bracket to -0.26 at the highest, relative to the lowest-income group. Wealthier students report substantially less anxiety overall.

However, the recession's *differential effect* (β_3) does not vary by income. Subgroup regressions across parental income brackets yield coefficients scattered close to zero with overlapping confidence intervals. Similarly, splitting by first-generation status reveals no significant interaction: β_3 for financial anxiety is 0.0013 (SE = 0.0015) for first-generation students and 0.0013 (SE = 0.0022) for continuing-generation students. For work expectations, β_3 is -0.0011 (SE = 0.0031) for first-generation students and -0.0067 (SE = 0.0029) for continuing-generation students.



Robustness Checks

Survey-weighted models and a narrow symmetric-window specification (2007 vs. 2009) confirm the baseline findings:

Specification	Financial Anxiety (β_3)	Work Expectation (β_3)
Baseline	-0.0011 (p = 0.421)	+0.0007 (p = 0.735)
Survey-weighted	-0.0010 (p = 0.596)	+0.0017 (p = 0.635)
Symmetric window (2007 vs. 2009)	+0.0016 (p = 0.399)	+0.0011 (p = 0.723)

Table 4: Robustness of the shock $\times$ post coefficient to survey weighting and a symmetric pre/post window.

7. Discussion

The central finding of this analysis is a robust null: state-level recession severity had no significant differential impact on entering freshmen's financial anxiety or work expectations. This holds across functional forms, event-study specifications, and demographic subgroups.

While the recession shock did not yield a differential effect across states, the baseline descriptive pattern remains clear: lower-income students experience significantly higher financial anxiety at college entry regardless of location or timing.

This continues to support targeted financial aid policy independent of business cycle movements.

8. Conclusion

Using a continuous state-level measure of Great Recession severity across nearly fifty states, this paper finds no significant differential effect on entering freshmen's financial anxiety or labor supply plans. The null result persists across five functional forms, an event study spanning 2003 to 2010, and subgroup breakdowns by income and first-generation status.

Future research could explore these dynamics using individual-level panel data (Yagan, 2019), examine county-level shock variation, or incorporate local job-market persistence metrics to better separate household wealth impacts from local employment opportunity.

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